# ActInf Livestream #019.2 ~ Deeply Felt Affect: The Emergence of Valence in Deep Active Inference

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hello and welcome to the active inference lab this is active inference live stream number 19.2 on april 13th 2021 so welcome everyone and thank you for joining we are the active inference lab we are a participatory online lab that is communicating learning and practicing active inference you can find us at the links that are listed here this is recorded in an archived live stream so please provide us with feedback so that we can be improving on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams today we're in the follow-up discussion 19.2 um our third discussion following on the dot zero contextualizing video and the dot one first discussion last week with casper and everyone else and so today in 19.2 we're going to be continuing this discussion on deeply felt affect hearing some slides from casper bringing up some of these questions that we're having about implications and other topics so really the goal is just to learn and discuss and we'll be going through whatever we need to and next week and beyond we'll be discussing different papers so let's just do a quick round of introduction and then we can see and hear what casper has brought to to share with us today so i'll start i'm daniel i'm a postdoc in california and i'll pass it to scott hi there folks i'm scott david i'm the director of the information risk and synthetic intelligence research initiative at the university of washington applied physics lab and i'll transfer to alex thanks i'm alex i'm researcher in systems management school in moscow russia and i pass it to blue good morning i am blue knight i'm an independent research consultant based out of new mexico and i will pass it to dean hi i'm dean i'm up here in calgary canada retired trying to keep up with all this active important stuff and i'll pass it to stephen hello i'm stephen i'm in toronto i'm just working on a practice-based phd i do a lot a lot of work with participatory theater and social topographies for exploring community development and i'll pass it over to i think it's now casper is that right yes so my name is kasper hasp i'm a phd student at the university of amsterdam focusing on the emergence of social cognition in development and evolution um and today i'll be will be continuing uh daniel said from last time so i'll be mostly assuming that you're already listening to that live stream as well but just for for everybody's sanity i'll give a short intro on what we've been discussing so far and just to describe the context of this research um i was i was when i started this particular topic deeply felt effect for this paper i was in the process of setting up um the topic of my phd research and focusing on social cognition but then in doing that um i realized that we were seriously lacking um there were serious gaps in the literature on how we actually connect from our bodily existence um to like abstract social world and our expectations about our social world shape the way we feel physically and how that recursively shapes the way we think about our social environment um and that's where i actually got into this kind of well you could call it a rabbit hole of affects and emotions um because i think that affect and is basically the kind of through emotions is what links um the social world the abstract social relations that we are you know experts at innately you could say even to a large extent if you look at the efficiency with which children in elementary school kind of track each other's popularity and whether they're part of the group or not in this kind of um i mean of course it's um we're not making i know like this dichotomy of nature and nurture it's an emergent system it's like there's an interaction going on um it's you can say that we're you know one of the things we teach each other to do um and that's where our system is kind of prepared to do in a large uh to large degree is to track events happening in our social environment and from an evolutionary perspective that's a very logical thing to do because for hundreds of thousands of years you could say our each of our yeah our survival would be entirely dependent on our capacity to be part of the group so to speak um our capacity to function in a group context not just on the individual level but also on the group level so our ability to survive as a species dependent on our capacity to coordinate and distribute tasks and specialize in different directions so uh in the very same way shaping and basically forming a body at a higher level of organization and just like the cells in our body are differentiated in different directions we as humans have to differentiate ourselves to form one large functioning body and you think like what does this have to do with the paper well um essentially the idea is that um the affective states as we talked about them last time in terms of kind of subjective fitness we focus largely on the affective states of a single agent doing some tasks in like a teammates but today we'll kind of move towards a steps beyond that and in particular the way in which tracking the effective states of others in in the body let's speak in the unit of organization as a group and how that's tracking can allow you to to find your place in the body just in a completely analogous to the way in which cells track each other's stress signals in the body so the pain um basically the empath our capacity to empathize with others would be very similar to the way in which cells respond to each other's stress signals uh yeah stephen i'll just be curious to know in your sort of journey often affect is tied to sensory feelings you know at the kind of personal level and so so that it becomes a kind of a a macro social sort of measure was that something that you already had in mind or is that something that sort evolved as you were exploring this these questions yeah i was essentially um interested in that connection between the very strong uh like experiential aspect of effect like when you feel embarrassed you're literally getting uh you can get a red face basically so it's like a very much an embodied aspect of that [Music] and somehow you're able to connect that to let's say what other people how um yeah your beliefs about how other people are perceiving you right so um that connection between the bodily states and those abstract um beliefs about the social environment is something that yeah has kind of driven this and this effort and scotland yeah one thing that's fascinating to me is the notion that there might be multiple levels of that harmonic coupling right and multiple pathways so that the dynamics of it it's certainly you know it everything every relationship and interaction is describable potentially as a binary and then then you can elaborate on it right and so there's is all sorts of nested affordances that have direct and indirect qualities to informing those interactions and so that picking apart and disambiguating the general relationship is fascinating right so that and the and it feels like just anticipating the surprises that active inference continuously brings us that there's going to be a i guess i would not just say a scale independence but also a local independence to some aspects of it so what i mean by that is they'll be analytical interoperability i would expect more broadly than we might get with other analytical framings simply by nature of the manner in which active inference approaches the questions of relationship so if you know what i guess i'm saying in that is i think it may be possible this is my years as a lawyer it may be possible to insert rhetorical artificial or let me use the word synthetic instead um linkages in there and that's something i'd be very interested in understanding what your thoughts are and how we can you know we assert them into being right and we and so describing them if it's efficacious to describe them in a certain way then they become adopted those things don't exist in nature until we describe them so we have an opportunity now to describe and disambiguate these things in a way that i'll use the it's like tempering with bach and the well-tempered clavier we get to decide on the notes or like when they decided on 360 degrees in a circle because it divides well i think that was the reason the phoenicians or whoever did that um so here we get to just maybe get to decide without doing an injustice to reality on how to parse the linkages anyway it's it's exciting because i love the way you you introduce that possibility of starting to pick apart those things and call and name them in a way that starts to do like david boehm stuff where every reality is first a thought and here we're having the thoughts so thank you for inviting us into that space it's fantastic thank you thanks for joining yeah um we'll get more into that in this second but i agree like it's kind of refers back to what i talked about last time in terms of compositionality or the kind of lego box in a sense that um yeah these abstract relationships can be applied on different levels of organization and that gives you a freedom in a sense that if you're able to capture this relationships on and the way they talk to each other on the different levels of description that's yeah that's really essential i think also blue was putting up her hands so so i'm anxious to see how this model unfolds in tracking others i think that that's a a great way to add an extra layer but i wanted to kind of ask first about uh the model as we discussed it last week it's just something i've been thinking about we talked about how um surprise is is related to like negative uh affects right so like you know whenever your prediction doesn't track that that's the the negative affects or or that's the as opposed to the positive uh valence and so i just was wondering like do we ever do it leave room for pleasant surprises like you know if i don't go food shopping i don't expect there to be a chocolate cake in my fridge but i come home and i find a chocolate cake i'm like this is awesome so is there like a room for that or or is it like incongruent or how do you resolve that yeah in fact it's a necessity that comes out of the fact that we talked about last time that the effective charge only emerges when there's a difference between the prior and the posterior so like uh you can be pleasant you can be positively surprised your predictions can be better than expected uh essentially so or the let's say reality can be closer to your preferences than you thought it was um or like further away let's say um so pleasant surprise is when you get a nice present from from your loved ones or something but then basically there's like a kind of um surprise in the sense that there's a certain preference being met that you hadn't anticipated um yeah it's like thanks thanks for the gift card oh wow it's so much it's like another surprise after you've renormalized it to the fact that you were receiving a gift you made an expectation oh maybe you'll be 20 and then it's more i mean there's that aspect and then there's um also purely on the information processing level so to speak when you're listening to music um or some kind of um something that's aesthetically pleasing there's often these kind of stimulation have some patterns that have inherent predictability so like they're you could say there's can be surprisingly easy to resolve um so it usually starts with setting up some tension and then suddenly resolving that you can see that in many art forms in storytelling and jokes telling jokes you basically have to create the tension and then suddenly the punchline resolves that tension and that's exactly the unsurprising or sorry the surprising part of that reduction in the free energy essentially listen but yeah that's a very good question i mean i think that's a very common misunderstanding also um that's yeah an understandable misunderstanding but that's because once you get to these hierarchical models things are often kind of uh get much more nuanced and you can't just say like surprise is always bad and it is surprised with respect to a particular generative model that specifies your ideal world so to speak that's that's what creates this negative effect but then once you on the meta level start to create expectations about how close you are to this ideal then once you get surprised on those expectations it can be negative or positive anyway so that's a long winded answer to your question let's have um a question from dave and then maybe casper will go through anything you want to present because we could definitely just keep on peppering with questions i totally feel it so dave and then continue on you're muted dave continue apropos of blues uh question some of the older researchers on affect notably sylvan tompkins and yak pong sepp go into a lot of detail on the differentiation of the emotions and sensations and affects and drives and so on one of the things that tompkins did a lot on was the speed of change very specifically uh a a rapid change in a sensation may have very different uh effect from a slow change also that's reflected in the face in the in what he considered inborn facial expressions underneath all the layers of acculturation a rapid smile is some drum is sometimes directly contradictory to a slow smile one a fast smile means oh you idiot a slow smile means oh you got it too a very good question i mean there's uh indeed the temporal dynamics they come into play are really um really important as well um and especially because these nested levels they tend to the higher levels tend to be slower so for feedback to trickle down let's say you need some time um so then if it's a slower kind of moving change then your top down let's say priors can already adapt a little bit before the next step happens um i mean that's what you see in for example in music where to really make something give to have an impact on the listener you need to have some kind of temporal slowness to the piece where there's like an intro and like different stages to really kind of prepare your expectations so to speak um or actually the joke example is also a good one like some people are bad at telling jokes they tell the punchline too early whereas like don't give the tension time to kind of emerge basically um yes sarah actually just made a comment in the chat um about how this model might be associated with humor and then can we think about humor in terms of surprise only or how do we think about these complex layers of emotion like catharsis surprise at different layers of the model like you had mentioned yeah i think humor of course many different levels um there's this really simple jokes right it's like one question and then and surprising answer i i sometimes in lectures i use this example like how do you call somebody without a body and without a nose nobody knows it's like a really silly one but it's kind of nice as an illustration because you can think basically the free energy is uh accuracy um sorry it's a complexity minus accuracy right so and when you get a question like that and you get two concepts without the body and god knows and you try to find one unifying concept and doesn't work and then you get the answer and the answer is both true false is accurate and it's like a reduction in complexity so you get like a double bonus in this case um but that's an easy one because the more interesting types of humor i guess emerge when you have like complex storylines and there are certain i don't know like satire and all these kinds of things so that's that's where it's not as easy to describe but i think it's also possible if you had some kind generative model of the social environment and then certain things are kind of inappropriate and then certain things are so inappropriate that they become funny or something like that like uh yeah there's every country has those comedians i think that they're able to to walk that thin line of uh extremity and so there was another question i think um can we do one more question oh yeah yeah yeah scott scott you want to go for it the question so much is just on on that on the measurement of the subtleties of the free energy there are books on rhetoric and i'm trying to find the title here but it specifically lists dozens hundreds of techniques of rhetoric like the juxtaposition of words in a sentence does one come before the other as priming or whatever a conceptual linking of multiple sentences paragraph structuring use of foreign languages french tends to indicate certain thing of style german tends to indicate accuracy whatever and so those notions of metrics from there are things that can be observed so they can be measured so the metrics from the humanities let's call it linguistic psychology they are positioned and ready to dive in to inform those variables in that space right and so that is the next interoperability is not going to be technical interoperability we got plenty of that it's going to keep going but what we have is this deluge of interactions as a result the true exponential increase in interaction volumes because of fifth order effective moore's law and what we need is meaning now to be interoperable and that's where we get to the drive here of the shared framings of rhetoric right and so one other point a hundred thousand years ago we were one village in the horn of east africa we radiated out around the world we came up with different food different language different politics different gods etc and now the internet brought us together for show-and-tell everyone's got everyone's work and active inference on each other whether we know it or not and if we name it as such it gives us a sandbox to play in at massive scales and i so i love the way you're naturally and i think appropriately and not by analogy but by direct application the concepts bringing in our behaviors that are beyond what are seen as base of maslow's hierarchy and needs basically it's beautiful you're doing beautiful work so thank you for inviting us into the space it's great thanks actually that comes back in a sec kind of needs hierarchy of needs or yeah how to deal with that so that's i i i just see one more question we can uh until we find a break time we can uh yeah are you yeah well mine isn't really a question in this i'm curious whether or not i have a question later but what i wanted to find out was when you discuss implications whether or not my question gets answered before i have to ask it because i think from an effective standpoint that's the part that's going to get really interesting so i just wanted to put that out there because i actually do have it written down because it's a long one but i want to ask it yet i want to see whether or not in the implications that you're going to share with us whether or not my expectations align with what you think we want to kind of find out more about oh see then yeah and i'll try my best to i mean it's impossible to be exhausted but uh i will try our best yeah um actually about the book the rhetoric of style i there's actually interesting another example this is like um the art of uh movie directing like choosing the shots and which shots are like kind of cognitively fluent so to speak and if you're creating a horror movie you like the exact opposite of that and the certain ways you can create anxiety or like different kinds of emotions based on how you how you move the camera around and which parts you do and don't show those kind of and one one reason to just this book in particular just advise that anyone take a look at this if they're looking at this it has hundreds of different eye examples each one's one page it's like sentence construction sentence basics etymological things tropes blah blah blah it just goes on it's like a checklist but it's this thick and so this is the things to measure for influencing other groups and so this is i really this is gene faunstock f-a-h-n-e-s-t-o-c-k i really thought this one was a really helpful one from a checklist and systems approach because we're really doing processes here trying to anticipate those processes that we can make into alcohol so good stuff um uh what was the title rhetoric or style someone's describing that rhetorical style the uses of language in persuasion by jeanne faunstock f-a-h-n-e-s-t-o-c-k and in fact one of the things just we're doing in our program is we're trying to come up we're not trying to come up with coming up with metrics for business operating legal technical and social variables so we take technically feasible systems like what we're talking about generally here ai and others and then test them for bolts reasonability and the way we get there is just by rhetoric because we it's storytelling it's folk ways you know they're just folk ways for the future and that's why the active inference gives us that framing it basically says you tell us what your things are in your minds you group and we'll take them and synthesize them and feed them back to you and then you get to decide whether you like them or not and so it's just really an interactive element so really thanks ella you're really interesting uh points yeah um all right so let's and let me try and get to the um the actual slides that i was gonna share but i'm sure we can find many of these uh slides um let's see how can i share my screen on the bottom left the middle button yeah yeah and then i'll resize it so it can look however it looks on your everybody's computer i'll make it look normal so no worries about that okay um looks good yeah okay so just to kind of recap a little bit but i mean we already already did that but last time we're talking about this nested set of um inferences essentially happens or appears to happen actually i'm pretty much uh to kind of hint at the instrumentalist discussion i i i'm pretty much fine with completely being completely and so that i don't i don't mind if it's um as i said yeah i i like to always say that all models are wrong but some are useful so i think this is a very useful way looking at it so it's instrumental whether it's true in a realist sense i i don't really care that much i guess in the end but mostly because i just don't have i i don't think there's a way to make that claim um and there's no way to prove it i mean there's a way you can believe in it or something but um and you can say that you require some amount of evidence to believe in it but as long as it works i i guess um for most applications that i'm interested in or maybe all applications doesn't really matter whether it's yeah what's what the philosopher thinks about whether it's real or not as long as it works um so that's my um i don't that's why i always say i don't have a dog in that fight um but yeah just to recap um a little bit too last time we talked a lot about valence but there's also other types or dimensions of affect and some of them are like uh there you are like three fundamental dimensions like arousal being like kind of your your activity um level your current amount of um resources that your your body is expanding and for the current um task at hand or current context and power control so the perception of whether out of agency you can say whether you are in control of the situation or not it's not exactly the same as valence in a sense last time we talked a lot about control as well but you can have a highly positive experience let's say if you're able to if your trusts that you're that your needs are being met even though you don't have control over them yourself um so there's these different dimensions that to some extent can can correlate with each other and but don't always and you have another one is something like approach and avoidance which is often related to valence but you can think of cases like um anger where you have an uh or aggression you have an approach reaction but you're angry um you're so you have negative valence high arousal um and an approach response so there's these different dimensions that sometimes correlate but not always and there are ways to let's say in very simplistic organisms there's yeah the dimensionalities very much reduced and when you're let's say a baby and you start out with very simplistic dimensions you feel content or you feel discomfort and as you get older you start to get more of this this dimensionality starts to um become more nuanced so to speak and it's hard to say and then the way um i'm relating this to to take this one step further um is to think about what this means for um when we have internal generative models what's that what those generated models and what kind of influence they have on your effective state your current affective state um and that's where it gets an interesting application in psychiatry because as we all know there is a dark side to our imagination and that is that imagination induced induced effect plays role in all kinds of mental conditions and also just on everyday life and so these are the extreme extremities of course the psychiatric diagnostic condition categories but everybody i think to some extent has experienced at least a few of these types of effects inducing thoughts and basically i would i can just run you quickly through them but imagination of imagined threats and flashbacks to some bad experiences in ptsd cravings so like imagining obtaining of drugs that you have that you're attached to but can be also other types of addiction ocd is obsessive disorder is a really really strong example people basically kind of you could say they're the effective responses that they get when they have certain thoughts are reinforcing the thoughts so they are afraid of breaking their own rules and they they get this kind of dread of breaking them so there's like a cycle and this kind of cycle is very self-reinforcing cycles of uh affective states and that then guide the way your imagination works and then that triggers more effective responses in depression and reaction i mean you can go on i think almost i haven't really found any um disorder in psychiatric um like the dsm that doesn't really fall into this except for perhaps they're really more like um physical um problems but those are often psychosomatic so and there's also a connection there um anyway so just for as a very abstract analogy we can construct a generative model to give this some flavor and can be of some social environment let's say you start in uh you have four locations um you can make this as complex as you want but just to give you to give for the sake of the simulation that i'm gonna be showing you can start neutral and so some safe space there is a place that's safe relatively safe but you get only a small reward and there is a place that's a little bit risky but you can get the high reward and why is it risky well because it has the probability of transitioning to this absorbing painful state you can think of this as like um asking somebody out on a date you have the risk of getting rejected or um as a hunter um hunting like a big big prey can be dangerous can get hurt and can also do something safer but you get a smaller reward so there's like many types of situations in the world where this kind of discount structure comes back where the higher higher rewards are often associated with potential and adverse events the problem with this type of state space is that it very quickly uh explodes when you want to search into the future and the events as they unfold and it's something we call combinatorial explosion but let's say you want to plan ahead and for a few time steps in this case four time steps you want to anticipate what's going to happen um very quickly it's gonna it's yeah the number of possibilities just becomes yeah hard to track and um the types of generative models that i've been working on and carl frisson recently uh with a paper with me and thomas barr and danny jar hofner um developed a new yeah i was basically sophisticated inference something i mentioned the last time as well um a way to do this kind of research without having the whole thing become computationally uh intractable and um yeah i'm not really going into the technical details here but essentially the core ideas that use instead of having the whole policy space which is multiplicative so you have in this case the policy space if for each transition you have three options here you have four so you get something to the order of four to the power four and a little bit less in this case but this is just for for um uh four sorry four to the power three um for just for three time steps and of course in reality we often anticipate um more steps into the future just yeah without even thinking about it explicitly so there needs to be a way to do this in a way yeah that basic that you prevent just overloading the system um with anticipating too many possible outcomes and what is interesting is that you can design a model in which the way you explore the future is guided by your effective states and this what i did for this simulation where you have this exploration of the future is itself an action model in a sense that you're experimenting internally with potential features and the way you uh trend yeah basically travel through the decision tree is guided by this action model and because it's an action model it's guided by your effective state and because it's the way i've linked that in my research is by adding having this action model precision um that we talked about in terms of the valence states um so what you get as an interesting example what i simulated here or so and model that can do overthinking can basically is something that you may have come across in the um when you discuss cascade effective inference a long time ago because then essentially this model uh action model precision is guiding the way the three searches is performed but then at some point this agent starts to imagine a little bit yeah some negative events and starts to um basically the precision starts to drop of that the way they explore in the and they start start to become more likely to imagine negative outcomes because when the precision on their action was high they basically had a confident kind of course of action in mind and where they would go and would automatically down regulate unlikely negative events but once this precision started dropping this agent start basically imagining these negative outcomes and that's made the precision drop even more and then this agent yeah you can see i kind of calculated in terms of the fraction of imagined events that were negative negative outcomes call it like threat perception or something and and it's just an illustration of how you could simulate a process something like what we could call rumination um or at least like yeah a negative consequences produced by overthinking the situation and that's interesting because it means that you're actually can also model too much and you would think in machine learning that the longer you run your your model the better it gets but once you get this kind of recursive relationship you can actually get a kind of breakdown in in the way the future is imagined or anticipated you bring this one step further you can also actually i have a picture here as well um this one so this is where um something that looks a little bit like muscles um hierarchy can come into play um even especially when you start to make this with an extra layers but here i just connected this sophisticate affective inference to higher level states that in this case that describe exhaustion hunger and valence state um and the point is and that's something i also highlighted last time it didn't show the model itself that and the more hungry you are the more strongly you're driven by preferences for fruits the more exhausted you are the more strongly you are yeah driven to prevent expanding energy and valence is then a kind of summary of how well you're able to manage all these different kinds of preferences and here i just show an example simulation where you have these three different higher level states and they're kind of valence er sorry exhausts the exhaustion and hunger are kind of competing in a sense because the agent needs to um to get foods they need to expend energy but yeah while they're at home resting their hunger increases the energy levels yeah the exhaustion decreases so there's this kind of cycle cyclical behavior where yeah if you follow it's kind of interesting you see that there is hungary in this case a very strong driver of negative affect so you see that whenever anger is high telling states is low um and the best situations yeah are in this kind of cases where you see that both of them are low at the same time so they're basically not hungry and not exhausted either i mean there's different kinds of balances you can implement here and this is in that sense just an illustration but you can already think of ways in which this also relates to things like depression for example where these uh people with depression would uh often report or have that had experienced depression and would report that they um have a lack of interest a reduced appetite because if you're really exhausted that means that you have a strong preference against expanding energy and that means that you can't anticipate the pleasure of eating anymore and you can't even experience it because you have this extreme and exhaustion or in this case the express yeah the depression states would be associated with experience of lack of energy i mean this is like a an example of how different kinds of phenomena that we know from psychiatric conditions actually emerge once you start to construct these types of models in detail let's see if i can so there's this aspect so this is kind of on the on an individual level and as i said i wanted share with you how this works when you start to think about multiple um unless there are any questions um i'll just continue you can hit um hide on that stop sharing thing to move it and then i think let's have dean and then steven so dean go for it well my question fits perfectly with what you just talked about so essentially uh comparing a t-search to a tree search which i think of as natural order and your your slide about combinational explosion my question was in simple likelihood mapping the binary is what it is the explorer action is limited answering what is good and what is bad on some version of valence continuum past that one wonders about what happens when trade-offs become trade-ins this would add a layer of complexity by keeping what is good what is bad and what if to the mix in the teammates example there would be a third option trap door with optioning as function being the key that their door would contain say four times reward and two times pain so not enough to cause permanent consequence but still exponential relative to the acute outcome there'd be no cue to this option only the barrier that is either charged or not is transparent you can see the reward you can see the lightning bolt signal signaling that the barrier could be charged based on what if which extends the epistemic and specific of the simplified likelihood mapping to now include distribution which is third possibility and perception sensing which is the potential shock just as part of a 4x result do you think effective charge remains continuous in terms of the path to renewed confidence i know that's a really long thing but i thought this was the direction you might go with this work and as it turned out way to go because this is exactly the kind of question when you have a learner who's thrown into a novel context they don't tend to only have a binary choice so how do we model for it because we want this to apply in real situations so clap clap that's perfect thanks i mean um the continuous versus discrete state space problem is pretty universal yeah um i think the way our minds tend to deal with it is by course grinding it's first in like categories and then later on when you're locally in the situation you end up sorting out the details on the continuous level exactly um so yeah that's i didn't i kind of ignored the continuous state story here and just for the sake of the simulation but what's nice is when you start to look at the parameters actually you get do get the continuous play or like the sorry the way the variables vary over the course of simulation you get the continuous part back so cool and also dean in that i i i saw like you're approaching a t-turn it's a discreet choice in the moment but when you pull back and when you include what if it almost uh makes continuous through the generative model of the learner in this case it makes that situation go from discrete to continuous and also when you're learning you could rest or you could eat and so it's actually like our affect as learners as we're optimally foraging it really does need to be considered in an embodied sense because at any point you know when reading casper's paper we all could have thought wow we're a little tired and then that changes that's a binary decision but it's part of multiple overlapping continuums so yeah it's reflect reconcile recovery cool so thanks for the question dean um stephen go for the question and then um anyone else and then if not casper yeah i'd just like to get your thoughts on how context comes in i see we think about scaling and it's very tempting to sort of all this can scale in many ways but this almost speaks to this idea of like they talk about trans contextuality i we need to think about different ways that we're in the environment and i think that the way that also ties in with the psychiatric challenges is that like you say the affect um starts to run away from itself so unlike someone who maybe doesn't have that disorder and can recalibrate or re-attune to the new environment someone with the disorder it's sort of like you say it runs away from itself but i'd be interested because one of the things with this work is it it puts the environment into the potential treatment regime and it doesn't just make it all about the mind and the body so just context and and how that that that has an impact um yeah so in these types of models in this case i simplified it but as we remember from the last time this d high level d would also include contextual states and then those contextual states can be inferred based on other things like external cues and so for example me observing myself um sitting at the dining table and smelling foods could trigger a kind of hunger state and then that increases my basically or like increase my preference for eating food so there's there is actually yeah there is space to include um just things like where the context determines the way your preferences are structured or at least influences the way they're structured so and this has then interesting implications for things like ptsd where people would get flashbacks when they're in a certain context that they associated with this experience for example or addiction as well it's like um people on the first line of treatment in strong addictions as just take people out of their contexts so that these cravings aren't triggered so and i think when you say trans contextuality that's a really interesting word i guess what you mean is that you have a different version of you for every different context in a sense yeah and it's also it's it's that that that that piece i mean they're talking about that a lot now with um developing um gregory bateson's work in terms of looking at multiple contexts and warm data they're talking about warm data so data it's not just dead data but is alive in a way the sort of stuff we talk about has affective content and trans contextuality means if you want to understand different contexts and not just deal with them separately and interrelate them then there needs to be some way that they're the living component of context is somehow held alive um yeah yeah that comes actually maybe that's a nice segue unless there are other questions there's a nice segue to the um culture part because great yep go for it yeah okay um because the most stable part of our context besides uh basic physical variables like temperature etc but our body temperature for example once you move beyond that and one of the most stable components is other people in our environment um and how we how we coordinate with them um and it comes back to also uh some of the points that uh were made on the rhetoric uh story but just on the i guess i kind of like to start bottom up so i just start with the way um the yeah the affected states translate um so shared affects and and cultured emotions so in this case um i i construct a few gerund models um i can have the ones that are associated with competition or with cooperation and there are different kinds of models and there would be um if i believe somebody is an opponent then i have a certain expectation about what they prefer if they're antagonistic they so in the sense that they prefer they like to observe that i don't i don't get any reward for example or they may be just competing in the sense that they just want to maximize their own gain and different kinds of generative models i can have for the way other agents behave so that would be like the the antagonistic and competitive versions and then there's like corporation um can also have different flavors you can have agents who genuinely have preferences for maximizing your gain as well or you can have agents who only cooperate when it just because it happens to maximize their gain these kinds of preference vectors is basically what you can infer based on how people behave and what kind of situations stay on um something like an altruist altruism versus egoism access you could think of it like that but and then beyond egoism there's something um machiavellian no it's something like you want the other person to have less so that's something i guess even beyond egoism and anyway there's different kinds of uh generative models that you can apply to make sense of the behavior of others um and just want to share with you a generative like a dynamic model that i made quite a long time ago before actually i got into active inference modeling per se and then it took a while to get it out published but this is work from 2000 2015. um and it was a pretty big challenge to get it published because people yeah it's like dynamical modeling is not very easy to sell to just psychologists who are used to their t-tests and everything but this model was about child play or actually play can be any interaction between agents actually it was child play was more like the immediate application but what is happening here is that every agent generates a decision and just the binary decision in this case um zero and one as was mentioned before you can often reduce it back to binaries um and in this case it's about playing so are they playing together yes or no if the agent tries to play with the other agent then only if both of them are trying to do that they have the emergent play outcome is playing together so there's a kind of conditionality embedded in their environment if one of them tries to play together and the other one doesn't then they then you get a zero outcome and since they both end up being low this kind of um builds in this dynamical model context i built different um self influences off on the cell this is based on a emotion theory by nico ryder i don't know i mean he's kind of famous in the netherlands emotional theorists but then kind of revamped for dynamical system theory and the basic idea is that you have these concerns so that was also referred to in this muscle pyramid um and they these concerns and tend to drive behavior but they also tend to drive or tend to generate appraisals in terms of whether your concerns are are being met and that can trigger emotional expressions which then influence the other person in the environment and vice versa of course so that was something from previous work but then yeah started to use this to generate um i mean yeah translate it to activate essentially um it's pretty easy to do so you can do this with almost any almost any dynamical model so if you're interested in active inference but you already are familiar with some other dynamical models usually it's a pretty straightforward translation it just means that you're reinterpreting certain parts of the mess sometimes means that you're reorganizing certain equations but often the core notions underlying model because i mean most yeah the models are always all i mean usually they're fought well through quite carefully by scientists so i tend to think of every model as a potential and yeah that has something of value that can be translated into the new into this interpretive framework and here it's about what kind of preferences the agents have and this is the lower level so i make it fix their preferences for being alone being together and then a kind of this is parameter here would be something like a symmetry bias so this is a preference against doing something else than the other person it's it's more of a hypothesis as i said before every model is a parameterization is a hypothesis it can be tested whether there's such a symmetry bias in practice there does seem to be such a thing anyway just not to overwhelm you with numbers but we can then make this hierarchical and let it play out over over time steps and then this agent starts to be able to manage their concerns over time so some agents might like to play together a lot more than others so then you get something like a baseline um concern for autonomy or playing alone versus a concern for um togetherness or belonging or something like that and those would be i think in every person are kind of competing so the idea here is that you're gathering evidence for each of these states and they are yeah they are driving your preferences but when you start to feel too lonely then you might be willing to sacrifice autonomy and when you start to um and vice versa when you need and feel you need more autonomy might be willing to play alone i mean this is a basic concept it can be applied to any kind of balance being act between preferences and you can also apply it actually i anticipate another slide here anyway i i've applied the same concept to model things like trust so trust games where you're trying to get a sense are we in a context where we're working together or are we in a context where we are against each other and it's actually the same types of moles but then with one additional twist is that now you're also inferring um yeah the the preferences of the other agent in a sense um with the same models applied so there's one i in addition to this there's one final thing i wanted to share let's see how much time oh we have quite some time business um i just wanted to share with you a few slides from to finish this from a talk i gave with maxwell ramstadt in canadian neuroscience spotlight was near in september somewhere in september just to go a little bit more in depth on what you do when you have this type of model that characterizes some psychiatric condition or could characterize it so there's always a different kinds of domains of the state space and and we start with something we kind presuppose a functioning uh or at least the potential to function or the model um and that's the kind of starting point but then you look at okay in what ways can it break down and what types of combinations of parameters result in different kinds of what we call computational phenotypes that could be associated with affective disorders in this case and what's nice there is you can now start to do computational phenotype together kind of map of different different kinds of problems that can emerge and this map doesn't need to be strictly categorical in the sense that the parameter space can be largely continuous so you can get clustering but it doesn't have to be absolute you can code mobility you kind of get it for free because there's like this kind of coupling and you can get yeah basically combine it with things like rdoc i don't know if you're familiar with that but it's kind of um alternative to the dsm which is a category as mostly a categorical approach our dog tries to focus more also in the underlying mechanisms that generate these different phenotypes and in that sense this type of computational phenotyping could be the underlying statistical tool that could guide um different kinds of investigations of not just affective disorders in the basic level but also social emotional functioning because of course as we started out in the beginning those are intricately linked to each other and that's where the context becomes important and the way this is the final slide that i want to share is that we can now use these types of tools let's say we can fit them to individuals and based on what a wide variety of data can be self reports can be objective measures be also things like even the more traditional measures could be used to kind of initialize the model and there's different kind of different approaches you can take there and the basic point is that now you can an um if you want to have a model of the model right it's it's a it's a metamodel it's a model of the model that tends to describe the way a person generates their behaviors and their faults and the way their feelings evolve over time once you have that you can use that to employ to basically forecast their weather conditions so to speak and those can be context dependent they can be too kind of tied back to a question that posed on trans contextuality we know that the chances of rain are dependent on certain contextual factors factors that don't really yeah that depends on high level dynamics on the scale that are not local yeah it's not local in the same way what's happening down inside your body or let's say your heart rate can be contextualized by what's happening in the rest of the system and in the end what's the kind of dream for this psychiatric application effectively of an effective inference would be that therapists and psychiatrists etc don't have like a static thing like a book like dsm that just puts people into boxes but instead they have this dynamic tool that um updates itself by through feedback with the clients then release that to help the therapists and the therapists can also improve that tool so in that sense you get a triad um where the and this can be applied not therapist client relationships but also can be applied to let's say help clients within their home environment to relate um yeah their loved ones their school and or their yeah their work environment this would be a personalized treatment prediction tool and treatment monitoring so there's a different levels of application i mean i focus here a lot on psychiatric applications because i think personally that's one of them yeah one that i find the most fascinating there's loads of other applications you can think of these types of models can be used for all kinds of commercial purposes if you think about entertainment industry how to generate an output that is kind of personalized um to entertain somebody i mean there's so many so many steps that you can take i just uh focused on the psychiatric side that i am yeah that's pretty much it for now but i'm further we can find some uh yep nice rabbit holes too let me see how i can stop sharing yes yeah awesome thank you casper super great to have you just kind of help us revisit the paper and also some of your past work and some of this other work and it really just struck me that you're talking about moving beyond the dsm into dynamic systems modeling the other dsm so it's a little acronym overlap but it just shows how when we have general principles it helps us descend into the nuance and the context the subtlety of a clinical relationship an entertainment relationship the educational relationship and actually do full justice to the complexity of the embodied situation rather than thinking that we're going to just have some sort of miso level theory that ends up actually falling in between sometimes general dynamical systems theory or complexity theory and then on the actual specifics so it was awesome to hear about these cases so we have um stephen and then scott and then anyone else who raises their hand yeah thanks this is really exciting with i've been i've been reading some of the work around the um the dynamical sort of options for psychiatry that you've been doing and i one thing i notice is you talk about the self the other and an outcome and like the outcome it seems as people are getting immersed in the outcome it gives a way to dissolve this self other dilemma which is in a way that the trap of modern psychology and psychiatry is you name someone the self because you can analyze the self and the other because you can name and analyze the other and then everything else just goes from there and the model is just like is it is dominant but never spoken about whereas here the model center stage but if you start to make it about the outcome and the context now suddenly the self and the other can start to take a more dissolved space you know and that and the positive psychology and the client can start to have more agency in their own life so i just yeah i'd be interested in just how you saw that relationship to self other an outcome yeah i mean in the in the interaction the distinction kind of dissolves to some extent in the sense that they are co-creating each other's models so to speak um and they end up having like a shared to a certain degree um and what makes this reality real or when like shared expectations is the fact that they have them uh together um and many of things in society are actually function only because other because we all to some extent believe in them like money is a very clear example like the only value money has is that other people believe it has value and again i mean self other i guess you can't really escape it in the end um i don't think we need to because if it's just a matter of the level of description i guess and just like the cells of the body are clear units but at the same time they are part of one hole and yeah this is like the gestalt basically this tall psychology kind of comes back to haunt you in a sense and there's the emergent hole that's other than the sum of the sparks i don't really mind talking about the units as long but i think what you describe is really important that you keep in mind that there is like it's just one level yeah in the end a reduction i think there was another is it's a follow-up question yeah well i think that makes sense i think just like you were saying if if the outcome is being measured does that change the way that we think about like if we start looking at the space the environment the outcome the context i mean are we in a whole different paradigm for treating disorders that's a really interesting one because i mean i know some really interesting research on the way in which psychiatric categories have influenced cognition and behavior across cultures basically people believing they have a depression or they have depression and then they can't get out of it because they ended up believing that that's their disease like a kind of discreet thing that they just stuck in so that's a really interesting example where in that sense are kind of like mind wear and they program and they kind of specify the way you structure your world and what i like about this kind of individualized modeling is that you can move away from those categories you don't even have to tell a person that they have a clinical diagnosis you can tell how you know how to help them because they have individualized predictions for treatment but you don't have to tell them like oh we're putting you in this bin there's somebody yeah that's like it becomes like a spectrum and there's attractors in that state space but we don't need to make these strict dichotomies anymore i mean maybe for the insurance that's the only limit it's like the insurance companies want to know do we have to pay for this or not it it's so true that the implicit story of diagnosis is you know you got two kinds of people the diagnosed and the undiagnosed and if you have three of these five symptoms you're in this box and it almost made me laugh at this last point about insurance how about uh pay per affective charge let's uh continuously tune that dial if they just need like you know a little bit pay a little bit or something there's a whole level of nuance that's when we're thinking about this in a in a big way so thanks for that answer um scott go ahead casper then scott it was something i also i was trying to i gave a talk in my psychology department or like the brain cognition meeting in my university whereas basically saying that in principle if we can alternate this we can treat everybody and to the extent that they're interested in like improving aspects of their life i think everybody has some things they would like to improve so there's like don't even need to just limit it to like the kind of model where you just wait until things go horribly wrong and then you start to try to help like preventive treatments that are not based on the sky and this kind of disease model essentially anyway and those prodromes might be um revealed through these inferred parameters like uncertainty so before the real situation begins it's really a good point so scott dave and blue so fantastic there are so many things so many threads here let me pull a couple together and it's on this space of social and internalized norms and folkways so dsm iv used to list the cultural phenomenon that weren't considered disorders like tarantella or you know the or the different things that you know they were cultural not considered but that always seemed to be very um imperialistic uh the way they did that but it's kind of interesting when you start to move into the um you mentioned insurance companies and what they will pay for the naming of disorders is is many of them are just call disorders because they don't let you function as an industrial worker you know what i mean there's all sorts of outside that dictates whether a disorder is a disorder and so i love that notion you're talking about about the clustering and the predictive elements there's an people in public health non-mental health physical disease health talk about forty percent four zero forty percent of the global health effort for any emerging pandemic is repeated from one to another so imagine the cost savings and resource savings if you start to do predictive mental health like people have been talking about the covet is causing the ptsd generation so you can have these clusters of things that need to be dealt with on scales of billions of people and that idea about merging treatment into norms so really not talking about treatment you're just talking about a continuity of existential support in in society that's that really is fascinating here's the question so it's kind of an interesting there was an article on soft paternalism a number of years ago and the economist that idea that for instance if i have a gambling problem i can register with the state of missouri to not let me into the gambling ships that go up and down the mississippi when i have a moment of clarity i can say don't let me on the ships and then i'm registered there and then they won't let me on the ships when i'm feeling the urge to go gamble so there's this kind of inviting that externality in one of the things i wanted to ask you about is your thoughts your gut thoughts on sovereignty and so i define a sovereign as something that doesn't need to ask for permission or forgiveness and it's always an external teleology it's always a projection out whether it's a deity or royalty or a country or a company we project it out so we can self bind to it okay that's my assertion okay so i'm looking it's that sovereignty has to have an abstraction of its subjects because that's the notion of so we project out it to be abstracting us so we can deal with the big problems and then the anomalies we take care with of with a mass customization of localizing it right so it feels like your clinical notion about psychological states could be employed for political purposes and i'm talking about in a positive way not in a bad manipulation but a good intentional manipulation of affective states that are positive for civic engagement and then you basically use the same notions as psychological disorders but you start correcting for let's call them political right because it's just normative states so that that's what i'm curious about is have you done any thinking about how to project cultural states from a group like like projecting it on the screen and then everyone watches it and says huh that's us up there let's go move towards that now sort of a question it's a really interesting question um i have a student that works on like communicative cultural dynamics assimilating the way in which beliefs kind of in a kind of minimalistic way that can be seen as um yeah modeled quite similarly to viral spread of um but then they're from the perspective of the beliefs that are spreading it's kind of like every time it's trying to be it's somebody tries to transmit it to the next person the question is how well does it fit with the next minds and only then only if their fit is close enough it can be it can reproduce essentially so then your question is kind of like one level beyond that where you're like we're trying to predict which message will fit to which group of people and then that can be taken in very equal directions but you can also kind of use it maybe to break um uh political bubbles and it leads to extremism something like that so there's different kinds of because in the end it's a tool um that's and one note on that political side because something people bring up is we're not simply talking about the communications or just the rhetoric display it's actually about doing a systems identification and potentially finding latent areas of agreement that are just two terms being used by two communities for so yeah but very important and topics i was just wondering uh sorry you talked about sovereignty so i was wondering how you view that connecting to sovereign because the way i see sovereignty is just like uh server and body is like something that has complete power over a certain domain and talking about people within that domain so they are completely under the sovereignty of that governing body right the way the way the link happens is my assertion is that all sovereigns are projections so they are they're they gods um i don't say we invented god because i'm an atheist but it might be a blasphemy to other people so we invented deities because there's different things you know so it's like we invented royalty we invented countries we invented companies name another sovereign they're the things that don't ask for permission or we invented them so that we can organize at larger levels now we didn't say it explicitly but that's why we did it and so all of them are projections so they're all stories from us now they take on a separate life sort of but they can never be identified that's the thing about sovereigns there's always a metaphysical element they can never be identical to the subjects that's why you that's why when a sovereign dies in office it's a it's a real difficulty for the sovereignty when a human who represents a sovereign dies because it has to be separated out you have a state funeral but then the body has to be separated from the body politic right it's hobbs it's a leviathan thing leviathan can't be composed of the individual parts it takes on a separate nature and anyway so that's why this is so interesting that projection and that act of collective projection is totally where we are right now think about all the problems climate change global resources water resources finance things that can't be done by nation states we need new sovereign levels active inference is going to be the way that we synthesize them it will be whether we call it active inference or not because it's all we've ever done as biological then living and social organisms and it's a way of framing it and so it's a way of now to me it's a new teleology we can project forward just like you're saying predicting on psychology i think we can predict politics and social states because we can use predictive analytics of clustering it's the same thing you take a bunch of americans another student who was modeling the elections american elections uh with the same um but i i like what you said about it's kind of subjective it's like the attribution of power um but then it's like a totally imaginary thing just like we attribute power to money yeah yeah i mean money is is de-risking consolidator you know it is one form of money wixel and smith and hume they all have different ideas about money but one way you can look at it is a de-risking consolidator so it takes all risk and makes it into one risk which is how can i get more money and if that's the case it's an interaction de-risking tool right actually this could give us a way to discuss um to bring it also to the earlier point of instrumentalism and realism because the fact that money has instrumental value doesn't make it i mean that's right that's like kind of interesting question that i don't have an answer to i mean i guess in practice it's real because people believe yeah it's like a queen singing and one thing one thing i just suggest you look at uh somebody just surprised me a couple of months ago about christopher alexander's work in architecture if you haven't run into that check it out because what he does is basically looks at patterns in architecture through deep time and and through space and the idea is that existing practices are captured in patterns of practice and if you want to look at a solution for an earlier for a problem look at patterns of practice from other situations and bring them into the situation here similarly it feels like we can in your work it's revealing that we can look at patterns of practice and then take them as predictive models and so that's what chris alexander does that in architecture and it's very straightforward there because it's physical instantiations like an archaeology of the mind of architecture in a way or the archaeology of the problem space anyway something to look at it might be uh fruitful thanks thank you scott so we have um dave blue and then a question from the chat so go ahead dave yeah um if you could uh turn toward the realistic side of your experience with with these do you kaspar have the feeling that people's real models are of the same structure as these transpersonal models or person to environment or organism to environment folks in this field seem to kind of uh assume that oh yeah it is all turtles all the way down or it's blankets all the way down and i don't know that i'm convinced that which what's your intuition interesting question um i think that it's kind of similar to the you can't last time we talked about dimensionality and you can project from 3d into 2d space but like understanding a 3d space while you're in a 2d space it's kind of impossible essentially in the end and it's a little bit similar like looking down on the hierarchy i think it's much easier makes things feel much realer more real because essentially they can they're within your yeah your mind can grasp them um and when you move upwards you're sort of the transpersonal part i think it's um our minds don't have yeah the capacity fully grasp something like that we can so it doesn't have the same experiential yeah i would just say detail maybe um so yeah i think you're getting at an interesting point where there seems to be a difference between the way what experientially feels real and what in it from an interme instrumentalist perspective and there doesn't seem to be much of a difference if you can move through the trans-personal level just like as i said i think in principle humans can form but higher levels of organization can form bodies together and every human is this unit in this body just like every cell is and every organ in the end um has its own local experience so to speak its own local data maybe that is being integrated to inform their states and their decisions but then there's the global pattern and that's the local elements never have direct access to it and i think that's something that gets at this point as you're describing um but yeah i guess i don't have the real answer just but i understand experientially what you're saying in the sense that definitely there are things that feel real there are things that don't exist but feel real that's also that there are ways in which trans personal concepts can start to trigger hallucinations in people for example and there's all kinds of hedge cases and does it answer your question i think it's a really interesting one i don't have a good answer it does yeah i think it is an elephant in the room perhaps um yeah and if i could just run off on a tangent on that um the relation between inner in the sense of what's inside a fristonian blanket and inner in the sense of what's going on in various parts of the nervous system or the organism i tend to come at this by thinking that what the the physiological uh the anatomical correlates of what we've been focusing on for the last two weeks are largely going on in the midbrain the brain stem some of the lower structures in the cortex but the modeling goes on in the neocortex so the inside is actually executed outside in the portion of the nervous system that's really good at doing algorithms and pattern matching and reinvestment of pattern transformations but the purpose of doing the whole thing is largely especially in a small child or a non-human mammal outside the cortex so that probably blows the whole thing in principle yeah from the models that i built it usually doesn't really matter where the data comes from like uh can be inter internal signals external it's more important what the inference is about so like um extraoception is is perception about external states of the world an imperception internal states of the system but i can extra external signals that may give me information about my internal states and vice versa so this distinction is pretty important but yeah that doesn't entirely resolve the issue just i like the fact that the models are kind of agnostic about where exactly the information comes from cool a lot to say on this um blue and then a question from the chat and then anyone else in the chat or here who wants to ask so i loved the diagnosis or the elimination of diagnosis and clustering instead of of diagnosing people except like when it comes to insurance and so casper i'm going to charge you with obliterating this in the minds of of insurance companies also because i think that there's this fundamental thing that happens and i mean maybe i'm wrong but if i am you know if i display some behavior of you know someone with adhd and i display a lot of those behaviors so if i display some of this behavior but then i go see a psychiatrist and i'm diagnosed with adhd i think that there's a large degree especially in younger children like as they're coming into themselves and dealing with identity crises teenagers and so forth i think that there's a large degree of of excusing or perpetuating my behavior because i have a diagnosis right so so if i'm diagnosed with something like adhd like maybe that will you know give me i'll give myself permission or excuse or or i'll cluster more strongly into the group of people that have that those kinds of behaviors because i have a diagnosis i think that this is very plausible what do you think yeah that's also what i was trying to get it in a sense that people tend to yeah use use these internal representations to structure their experience and then to respond yeah to generate their responses and then these boxes become self-fulfilling prophecies essentially um to a certain extent then there's that side but there's also then there's another side to it is that there are also positive reports that people when they can attribute the effects to something it makes them feel less anxious so they can kind of explain the way and yeah i don't know what's yeah it has downsides and upsides i guess yep a very sensitive and interesting thing but it reminds me of the 12 steps and the first step is to acknowledge the kind of person that one is obviously we're not going to go into this and it's for others to investigate but it's a trade-off between having a narrative certainty a diagnostic confidence versus the alternative and so there's our teammates in meaning or a narrative space which one of these bifurcation paths narratively will we then have the lower levels of our model like whether we exercise or not or whether we have a job or not those kinds of things come into play um i just want to ask the question from the chat and then um dean and stephen and scott so in the chat someone wrote can free energy principle based models reach a level in the future to be able to precisely predict when emotion or valence happens and more importantly when in the temporal dynamics of that cycle does thought subside so how is our experience of thought related to some of these simulations with uh you know the countervailing affect of chart or countervailing you know hidden states where does our experience of thought come into play i mean um it's to some extent already being done in the sense that um there's applications of in amsterdam they have uh they're trying they have been treating ocd with the brain stimulation and they identify the locus of the intervention using fmri and fmri is driven by dcm which is driven by free energy minimization right so we're already doing that to some degree identifying the kind of recurrent self-destructive loops of behavior in in the brain so internal behavior of the system and then identifying ways to break break those loops so they they implant they just try to find the kind of note at which the loop is um through which the loop is circulating and then they break yeah they disturb that note once in a while and actually leads to an improvement in function it's kind of funny because when they remove it and loops tend to return pretty quickly again so seems not to be an it's it's kind of it's not a pool treatment it's more like prevention it gets more interesting if you had like real time way of getting into that then you don't yeah you would need not just like um an electrodes in there but like a whole a whole array of electrodes that are like tracking and it's a little bit scary but in principle you could have like a whole weather tracking system inside and then um if there's a hurricane emerging and blocking the i mean they tried to do that already for people with epilepsy i guess then we all become like a cyborg or something if you start to go that and that's what maybe not i mean what i guess i find more interesting or like more appealing is um if we start to become more aware of how these feelings generated by our relation with the external world so there's this use case of um tracking um warehouse workers that i've been uh seeking to apply some of my work to where you could track their affective states over the course of the day and every person has a different chronotype every person has different times of day when they feel most energetic and it depends also on how well they've slept recently et cetera and so if you had some intelligent warehousing system then you could base you could distribute tasks and the schedule of the warehouse could be adjusted to suit the preferences and to maximize well-being of the the workers and potentially in the process also increase productivity so that's like kind of capitalistic potentially if we could construct things like that and we could use our our modeling of these effective dynamics over the course day by day for each individual we could use them to care for people to make them to prevent burnouts and potentially make that appealing for people who are just driven by money by profit unfortunately they're always in the end just most companies are just driven by profit so to make that realistic and scalable you need to show that also is good for the for the business um anyway so that's like some more parts i have in reaction to that question um there was a sub component what was the second part could you read that again um uh more importantly where in this temporal dynamical cycle that's a very nice question i wish i knew let's keep let's keep clicking on it i guess there are certain parts of our system that are able to disable are kind of contributing to our verbalization and the way i feel thought at least the kinds of introspective thoughts that uh it's verbal um tends to be a kind of internal internalized social skill essentially you first need to learn about how to speak with others and then as a child i guess when you're really young at some point you start to be able have these dynamics run internally and my suspicion is i think there are some research actually confirming this to some extent but in order to create the verbal thoughts you need to access different pathways you just attenuate the actual motor execution um that's how they tend to think about it in the literature um and then there's imagery thought which is yeah in a sense even more amazing maybe because that's not something you can i mean verbal uh that's an actual action that you do so it's kind of easier to imagine how you could have this action evolve like unfold internally but imagery is not something that you do in life it's just something you can you control your own internal pathways that you normally you would use for for having a visual sensation your mental actions can somehow control your visual processing pathways which i i have a hard time imagining how exactly that i mean i can build a model of that i think but yeah how it works in like biologically i find it quite hard well that's that's the instrumentalist tango we can build a few model architectures we could think about some differentiating experiments we could imagine some informative observations go from there continue the discussion from there so um in our last little bit let's go dean scott stephen thanks uh real quick um you talked about personalized stuff and the loop seems to come back on politics because i remember a couple weeks ago the first thing i think i said to you after high was we can't unlearn this once we start using these diagnostic tools we can't unsee it um so there's a lot of politics involved and people have this ability to to see things one thing i want one reference i wanted to mention and be especially i think helpful to scott if he hasn't already read it about this personalized versus political is an author by the name of paul khan paul w khan and his book is called origins of order and what it does is it tries to bring those two ends of that continuum into some sort of a not like there's no balance between the two but there's kind of a back and forth and it's very looking at rules but not on a legal level but on the on on much bigger scales so it kind of fits nicely with what blue said about the diagnostics what we've touched on a couple times about the instrumental stuff but i still think that if you try to apply a lot of this stuff to which is what i did before it would i used very rudimentary ways of expressing this but i gave young learners these tools and boy was it political the results of that because they could just see things and start doing things and there was a whole lot of people that went hey whoa whoa whoa they're getting way too much power from this but anyway that's just an aside i just wanted to share the reference because i think it really ties in nicely with some of the stuff that scott was talking about earlier and obviously i don't think the political part of it goes away no matter how much we want to be agnostic thanks i mean yeah in a political context in a sense what we're trying to determine is what what is the generative model that governs the body so like this conflict and tension and tension resolution and it's kind of like the price we have paid for being able to to organize it's it's the body politic the leviathan as brought up earlier will we turn left will we turn right will we look forward will we look backwards there's a lot to this multi-level political approach so scott and then stephen yeah thanks so the it it is a big um it's like moving a slime mold but you know from outside the um a couple of things that jumped in in the last bit um and and where i'm going with this is the notion of a capitalist enclosure of consciousness with the categories of abstraction so the painter kandinsky and i've said this before it just praised on my mind painter kandinsky said that violent societies yield abstract art and one of the things i've always wondered about is is the reverse also true is abstraction a form of violence and so here one of the things that i was thinking about in the last couple of comments was symbolic synesthesia right we don't really have a perceptual synesthesia but we take symbols and we substitute them in for things so like if i drank tequila when i had a bad experience with something else a bad tequila experience with something else then i'll remember the bad tequila experience as associated with that other not that i ever had a bad tequila experience i'm just saying if it never happened so that abstraction you know we had it kind of becomes our prior whatever and it goes back to that notion of the the humor element too the abstractions are taking us to a place and then revealing the reveal comes right and so one of the things i was wondering about on the capitalist side we don't want surprise we don't want punch lines for jokes because that's risk right now now you don't have innovation then that's the problem is there's this tension between innovation and stasis right and so they want innovation but they don't want somebody else to innovate and take the risk but then they want the benefit but anyway so from a capitalist enclosure perspective one of the things i'm wondering is does active inference and this is the does active inference we we're all taking care of each other's active inference exhaust right we're all taking the where everyone's doing active inference on each other and we're all the externalities for everyone else and so we're all having to internalize everyone else's discovery pathway in a sense right we do we're loading on each other so that of exhaust management is a commons in a way we're co-managing risk so when i talk about money as a universal risk consolidator might the metrics of value in markets give us insights not entire insights because markets don't tell you everything not everything is measurable that way but give its insights into preferences in a way that starts to give us the insight into that co-management of the active inference exhaust essentially so it's really treating commons not as a fisheries commons like with an asset commons but as a risk commons and it's basically consciousness as a so the imagine that the mind exists in language rather than the brain the brains are tuned into the shared mind and now we got to manage that share mind so we get nonsense in there something that's wrong science or bad politics or not functional economics or whatever and we want to get rid of that because it's spoiling the commons so just that was what came to me that this idea the sovereign feels like how we synthesize an externality to say whoa whoa whoa we got to get this right because we got some big scale stuff we got to do like ants and bees can do without memos we gotta we have memos so we got to write the memos so what do the memos say and that's why you get to dsm and stuff like that and what they're saying is just like foucault said pastoral control is what drives people together with certain behavior right and so it's that same thing are we being driven you know is it is are we being driven to be industrial workers or now we're being driven to something else and how do we manage that that commons of our infants active commons to aspire to what we want to be now there's a question sorry i mean it kind of makes me think about the way in which um the major religions have mostly emerged as soon as um at least during the same amount of time that they ended up being able to write it down basically so it's like a pro it was a recursive relation of course but you basically need a culture um for that level of unity there's two directions you can go and i guess it's probably both ways but on the one hand you need to be organized enough to have a culture that you're able to write things down and once you're able to do so you can basically you can kind of hard code the structures and you end up having a competitive advantage over other potential models essentially so maybe the mind and culture are emerging phenomenon of complexity right maybe complexity causes the mind because we need we need to when we say we need to do something we often think it's in response to something but maybe that thing itself is another instantiation of complexity itself and so maybe what we're doing is merely red queening the whole thing right just keeping up we think we're actually innovating and doing things but really what's happening is with increasing exponential increase in interaction volume and this and this is not as this is what i actually believe in pursuing if there's a raw increase since the big bang in exponential increase in interaction volume maybe all we do is just keep up and keep um dealing with that externality as living organisms in order to maintain neg entropy states maybe we just have to keep rushing into higher levels of order organization in order just to not dissipate anyway thank you scott there's also opposite ways thinking about like um where these organized structures are like the best way to dissipate energy so there's like a kind of just like the convective bubbles in uh in some kind of uh heated um volume of the water or fluid and the substructure basically helps the system to dissipate the energy um so there are certain it's like interesting two ways of looking it's the same thing in a sense it's that structural dissipation of risk is what you have in law and that's that's all it is it's just it's a time machine that predicts the future everyone can look at the prediction in a contract or law it's enforceable and they can rely on it and so it saves resources so it's exactly what you're saying thank you um stephen if you want to make a really quick last comment because we're out of time and then we'll close out yeah thanks um i was just going to say i mean sort of ties in this idea of abstraction and the um is i do a lot of work with action methods and ways of working in space and i think it ties into this question i think it might also tie into next week's talk is there's certain things that we can realistically get a handle on if we can take a perspective on it so for instance you talked about exteriorception that sometimes it gives you information you take a perspective on to reflect on your internal states and maybe you look at your internal states but i think there is this danger that this abstraction starts to take over because there's a whole load of levels particularly you know when it's the embodi which we can't see and that's okay you know these these blankets are blankets can go down and um it's beyond our perception to take a a thought on and the reason that i think that can be interesting is um is because in in action methods and psychodrama and these types of action approaches rather than the freudian approaches which basically took over is okay you can't take all these percep perspectives and analyze it in the way that we've got in this habit of doing but these other ways of knowing embodying exploring i think that that also is another direction which often has not thought about so i just think that um as we go into this what you're saying and the way that we think about instrumentalization is that you know we might need to instrumentalize using the the sort of the the more embodied effective methods and processes as well as this kind of other approaches because i think that we will hit the limits of an analysis and i think that might be something that this conversation sort of talked about so anyway i just thought i'd put that there if you've got any thoughts on that and i'll thanks a lot for a great presentation yeah i mean i think you make a very good point and that's uh and thanks for the compliment and i mean there's the the connection or the integration across levels let's say that's important for your well-being then what you describe is a very logical pathway to improve people's mental state in the end as well when they're more connected with their existence instead of living in that abstract space um i mean i think i have as a scientist i have a probably increased risk of going down that path because you're always thinking about things so yeah um i totally agree that that's probably many treatments actually i mean that's why meditation is such a appears to be socially effective as well anyway thanks for the thoughts yeah very good point yeah thanks so much everyone for participating and casper it's you're not always thinking sometimes you're coming on and hanging out with us i guess but maybe thinking during that too but really um fun discussion we hope to have all of you on again whether it's a paper you've authored or not so thanks a lot and uh yeah until next time